

Adversarial attacks

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1 Introduction

Since Szegedy et al. (2013)’s findings on robustness issue of modern neural network, there has been a huge chunk of literature discussing adversarial attacks and corresponding defenses. These attacks are usually imperceptible to eyes, but are actually deliberately perturbed versions of original inputs. For example, in Figure 1, by adding a small perturbation on a pandas image, the adversarial attack successfully fooled the classifier into regarding it as gibbon, with a very high confidence. Such perturbed inputs are so-called adversarial examples and can be used to attack a trained model, which might be a severe security threat to many applications such as face recognition and semantic analysis.

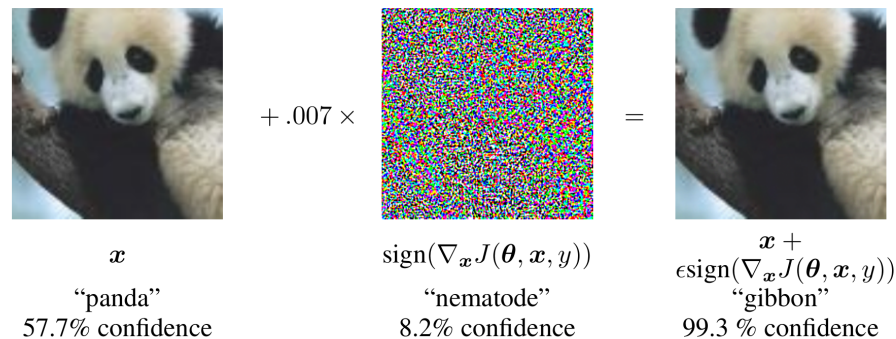


Figure 1: Apply fast gradient sign method to GoogLeNet on a pandas picture from ImageNet (Credit: Goodfellow et al. (2014))

There are some questions of interest:

1. Why there exists such adversarial examples and how ‘dense’ they are?
2. How to mitigate the influence of adversarial attacks and stabilize the trained model?
3. How to detect whether an input is adversarial or not?

2 Attack

The earliest idea and widely-used attack method is directly searching an adversarial example whose distance to the original data point is within the noise level. The key optimization problem is

$$\max_{x' \in B_\epsilon(x)} \|f(x') - f(x)\|,$$

where $B_\epsilon(x)$ is an ϵ -neighborhood of x . Since this optimization problem itself is intractable, (Szegedy et al., 2013) proposed using box-constrained L-BFGS to find an approximation solution, while (Goodfellow et al., 2014) proposed using a gradient-based approximation named fast gradient sign method

$$x' = x + \epsilon \operatorname{sign} \nabla_w f(w).$$

This gradient-based algorithm is computationally feasible and fast, so a number of follow-up variants are constructed based on this idea (Madry et al., 2017). The second-order approximation by semi-definite programming is also studied (Raghunathan et al., 2018b).

Finally, an interesting observation is that many researchers notice that the same adversarial perturbation applied to different inputs will result in wrong classification, and even fool different models (Moosavi-Dezfooli et al., 2017). In other words, adversarial perturbations are somewhat ‘universal’ and ‘transferable’.

3 Defense

To defend against adversarial attacks, there are many tools at hand. For example, the most natural choice would be to feed adversarial examples into neural network during training. This can be thought of as data augmentation technique. However, this naive defending mechanism can be exhaustive in computation, and may be further deteriorated by other attacks. An alternative approach is to modify the input during testing. For example, Cohen et al. (2019) proposed to smoothing over predictions from perturbed test images, resulting in a “majority vote” prediction, which can achieve provable robustness within an ℓ_2 ball. Another direction of defense aims at controlling variations implied in the network structure. For example, by restricting global Lipschitz constant of the network, we can guarantee robustness against small perturbations (Szegedy et al., 2013). Also, Raghunathan et al. (2018a,b) proposed to control the adversarial loss in the worst scenario via semi-definite programming, which can provide certifiable robustness against small perturbations. Note that defenses based on controlling Lipschitz constant and worst-case adversarial loss can be conservative depending on tightness of the upper bound.

References

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